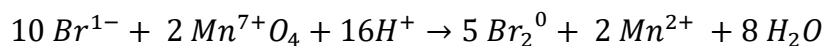
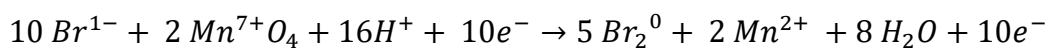
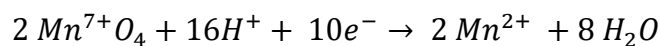
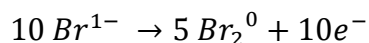
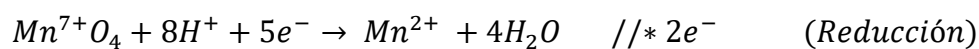
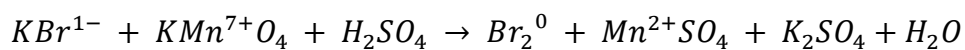
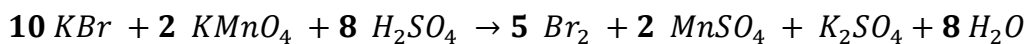


RESOLUCIÓN

1.-



Tenemos:



a)

Agente Oxidante: $KMnO_4$: 2

Agente Reductor: KBr : 10

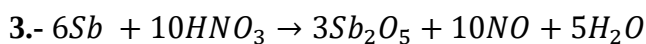
Sustancia Reducida: $KMnO_4$: 2

$$X = \frac{Ag. Reductor - Ag. Oxidante}{Sustancia Reducida} = \frac{10 - 2}{2} = \frac{8}{2} = 4$$

b) Coeficiente del Sulfato de Manganeso, $MnSO_4$.: 2

2.-

$$250 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}} \times \frac{1,5 \text{ mol NaOH}}{1 \text{ L}} \times \frac{40 \text{ g NaOH}}{1 \text{ mol NaOH}} = 15 \text{ g NaOH}$$



$$51,4g \text{ Sb}$$

$$120 \text{ mL } HNO_3 \text{ 6M}$$

$$V_{NO} = ? \{25^\circ C, 730 \text{ torr y } P_v \text{ de } 24 \text{ torr}\}$$

$$51,4g \text{ Sb} \times \frac{1 \text{ mol Sb}}{121,7 \text{ g Sb}} = \frac{0,422}{6} = 0,07 \rightarrow \text{Reactivo limitante}$$

$$120mL \times \frac{6 \text{ mol } HNO_3}{1000 \text{ mL}} = \frac{0,72}{10} = 0,0072 \rightarrow \text{Reactivo excedente}$$

$$P_g = 730 - 24 = 706 \text{ torr}$$

$$0,422 \text{ mol Sb} \times \frac{10 \text{ mol NO}}{6 \text{ mol Sb}} = 0,703 \text{ mol NO}$$

$$V_{NO} = \frac{(0,703)(62,4)(298)}{706}$$

$$V_{NO} = 18,4 \text{ L}$$

4.-

$$\frac{1,84 \text{ g}}{ml} * \frac{40 \text{ g } H_2SO_4}{100 \text{ g}} * \frac{1 \text{ mol } H_2SO_4}{98 \text{ g } H_2SO_4} * \frac{1000 \text{ mL}}{1 \text{ L}} = 7,51 \text{ M}$$

5.-



$$8,7 \text{ g MnO}_2 * \frac{1 \text{ mol MnO}_2}{87 \text{ g MnO}_2} = \frac{0,1 \text{ mol MnO}_2}{1 \text{ mol MnO}_2} = 0,1 \text{ Reactivo limitante}$$

$$200 \text{ ml solución} * \frac{3 \text{ mol HCl}}{1000 \text{ ml solución}} = \frac{0,6 \text{ mol HCl}}{4 \text{ mol MnO}_2} = 0,15 \text{ Reactivo en exceso}$$

$$8,7 \text{ g MnO}_2 * \frac{1 \text{ mol MnO}_2}{87 \text{ g MnO}_2} = \frac{1 \text{ mol Cl}_2}{1 \text{ mol MnO}_2} * \frac{22,4 \text{ L Cl}_2}{1 \text{ mol Cl}_2} = 2,24 \text{ Litros de Cl}_2$$

Respuesta: c) 2,24 litros